



ACTIVITY 06 -SANKEY PLOT

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COURSE: DSA0601-Data Handling and Visualization

A university wants to analyse the career pathways of students after graduation. The available dataset records the flow of students from their academic departments to specializations and finally to their career outcomes.

The dataset contains the following hierarchy:

- **Departments:** Computer Science, Information Technology, Electronics, Mechanical, Civil
- **Specializations:** AI & ML, Data Science, Cyber Security, Embedded Systems, Manufacturing, Construction
- **Career Outcomes:** Software Engineer, Data Analyst, Research, Higher Studies

The relationships between these categories are stored as source node, target node, and flow value, where the flow value represents the number of students moving from one stage to the next.

Using R and the networkD3 package, answer the following:

1. Create a Sankey Plot representing the complete flow of students from departments to specializations and career outcomes.
2. Configure the diagram with:
 - Node width = 35
 - Font size = 14
 - Career outcomes displayed on the right side of the diagram.
3. Interpret the visualization and identify:
 - The specialization receiving the highest number of students.
 - The career outcome with the largest overall inflow.
 - Which department contributes the most students to Software Engineer positions.

4. Explain why a Sankey diagram is more appropriate than a bar chart or pie chart for visualizing this dataset.
5. Suggest two improvements that would make the visualization more informative for university administrators.

Expected Output:

A Sankey diagram showing the flow of students from Departments → Specializations → Career Outcomes, along with a brief interpretation of the major pathways and student distribution.

Database:

ID	Node
0	Computer Science
1	Information Technology
2	Electronics
3	Mechanical
4	Civil
5	AI & ML
6	Data Science
7	Cyber Security
8	Embedded Systems
9	Manufacturing
10	Construction
11	Software Engineer
12	Data Analyst
13	Research
14	Higher Studies

Link table:

Source	Target	Value
0	5	80
0	6	70
0	7	50
1	6	60
1	7	50
2	8	90
3	9	100
4	10	80
5	11	50
5	13	30
6	11	40
6	12	60
6	14	30
7	11	35
7	13	25
8	13	40
8	14	50
9	11	70
10	11	50
10	14	30

Explanation:

1. Data Science gets the highest number of students, with 130 students.
2. Software Engineer is the most common career outcome, with 245 students.

3. Mechanical contributes the most students to Software Engineer jobs, with 70 students.
4. A Sankey diagram clearly shows the flow of students from departments to specializations and then to careers.
5. The diagram can be improved by adding different colors and interactive labels to make the information easier to understand.

Program:

```
# Install package - run once
install.packages("networkD3", repos = "https://cloud.r-project.org")

# Load package
library(networkD3)

# Create nodes
nodes <- data.frame(
  name = c(
    "Computer Science",
    "Information Technology",
    "Electronics",
    "Mechanical",
    "Civil",
    "AI & ML",
    "Data Science",
    "Cyber Security",
    "Embedded Systems",
    "Manufacturing",
    "Construction",
    "Software Engineer",
    "Data Analyst",
    "Research",
    "Higher Studies"
  )
)

links <- data.frame(
```

```

source = c(
  0, 0, 0, 1, 1, 2, 3, 4,
  5, 5, 6, 6, 6, 7, 7, 8, 8, 9, 10, 10
),
target = c(
  5, 6, 7, 6, 7, 8, 9, 10,
  11, 13, 11, 12, 14, 11, 13, 13, 14, 11, 11, 14
),
value = c(
  80, 70, 50, 60, 50, 90, 100, 80,
  50, 30, 40, 60, 30, 35, 25, 40, 50, 70, 50, 30
)
)
)
# Create Sankey diagram
sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
  nodeWidth = 35,
  fontSize = 14,
  nodePadding = 15,
  sinksRight = TRUE,
  units = "Students",
  width = "100%",
  height = 650
)

```

Output:

